CLEMENTE GOTELLI

Hydraulic Engineer and Researcher

Lausanne, Switzerland | +41 78 205 9383 | clemente.gotelli@epfl.ch | https://www.linkedin.com/in/cmgotelli

PROFILE

Hydraulic engineer and EPFL-trained researcher with experience in flood protection, water-security engineering, sediment transport, and numerical modelling. My work has addressed water-related climate risks through flood-protection assessment under climate-change scenarios, drought and extreme-rainfall risk assessment, drinking-water source protection, and Alpine natural hazards. I have a strong technical foundation in hydraulics and modelling, combined with experience in applied research, public-infrastructure evaluation, and teaching. I am interested in contributing this perspective to collaborative work on hydrological risk, climate adaptation, and societal decision-making.

RELEVANT EXPERTISE

Hydrological and risk focus: Floods and flood protection; drought; water security; mountain hazards.

Methods and technical tools: Hydrological and hydraulic modelling; numerical / CFD modelling; risk assessment; flood-scenario evaluation.

RESEARCH AND PROFESSIONAL EXPERIENCE

Postdoctoral Researcher **[09 2025 - Present]**

EPFL STREEM Lab | Lausanne, Switzerland

- Developing an open-source numerical model for avalanche-induced lake tsunamis, supporting the assessment of cascading hazards for Alpine reservoirs and hydropower infrastructure under changing cryospheric conditions.
- Applies hydraulic and numerical-modelling expertise to natural-hazard questions with direct relevance to risk management and climate adaptation in Switzerland.
- Research scope also includes river morphodynamics, sediment dynamics and open-source CFD modelling.

Doctoral Researcher and Teaching Assistant **[11 2020 - 08 2025]**

EPFL, Environmental Hydraulics Laboratory | Lausanne, Switzerland

- Conducted doctoral research on geophysical flows in mountainous environments, with numerical-modelling expertise relevant to geomorphological processes.
- Teaching assistant for Hydrological Risks and Land-Use Planning, and for hydraulics and fluid-mechanics teaching; supported the explanation of technical risk concepts to environmental-engineering students.

Hydraulic Engineer **[08 2019 - 10 2020]**

CDM Smith | Santiago, Chile

- Contributed to the design and assessment of river and flood-protection works for high-flow conditions.
- Worked on turbidity prevention and early-warning studies for the drinking-water system serving Santiago, linking source-water hazards to water-supply continuity.
- Contributed to desalination projects in Chile and internationally, supporting water-security solutions for populations and industrial users.

Researcher **[07 2018 - 07 2019]**

CIGIDEN - Research Center for Integrated Disaster Risk Management | Chile

- Worked in an interdisciplinary disaster-risk research setting where hydrometeorological extremes were examined alongside their societal and infrastructure implications.
- Contributed to disaster-risk assessment projects focused on drought and extreme convective rainfall in Chile.

Applied Project Evaluation **During MSc studies [2018]**

Chilean Ministry of Public Works, Social Evaluation of Projects division | Chile

- Designed and evaluated flood-protection projects under climate-change and extreme-value scenarios, including assessment of net social benefits for public investment decisions.

EDUCATION

PhD **[2025]**
EPFL | Lausanne, Switzerland
Doctoral research in environmental hydraulics and geophysical flows in mountainous environments.

MSc in Hydraulic Engineering **[2018]**
Pontificia Universidad Catolica de Chile | Santiago, Chile
Relevant coursework: Hydrology; Hydrological Modelling; Urban Hydrology; Hydrology and Extreme Events.

Bachelor's degree **[2016]**
Pontificia Universidad Catolica de Chile | Santiago, Chile

LANGUAGES AND ADDITIONAL INFORMATION

Spanish (native) | English (C1) | French (B2) | Italian (A2)

Cross-sector experience at the interface of academic research, engineering practice, public-infrastructure evaluation, water management and teaching. Highly committed to collaborative work across hydrology, climate risk and societal decision-making.

PUBLICATIONS

- **Gotelli, C.** & Ancey, C. Morphological State Transitions and Sediment Transport Variability in Braided Rivers. Accepted in Water Resources Research.
- **Gotelli, C.** & Ancey, C. Braided Rivers as Morphological States: A Data-Driven Identification of Recurrent Planform Configurations. Under Review in Water Resources Research.
- **Gotelli, C.** & Ancey, C. Bedform evolution simulations using continuum-based models and stochastic transport equations. Manuscript in preparation to be submitted to Journal of Geophysical Research Earth Surface.
- **Gotelli, C.**; Sandoval, J. & Musa, M. Interaction Between Hydrokinetic Turbine Wakes and Bed-Induced Flow Structures. Manuscript in preparation to be submitted to Renewable Energy.
- Sandoval, J.; **Gotelli, C.** & Escauriaza, C. Vortical dynamics of supercritical flows in the vicinity of in-stream obstacles. Manuscript in preparation.
- Sandoval, J., Soto-Rivas, K., **Gotelli, C.**, & Escauriaza, C. (2021). Modeling the wake dynamics of a marine hydrokinetic turbine using different actuator representations. Ocean Engineering, 222, 108584.
- **Gotelli, C.**, Musa, M., Guala, M., & Escauriaza, C. (2019). Experimental and numerical investigation of wake interactions of marine hydrokinetic turbines. Energies, 12(16), 3188.

REFERENCES

- **Christophe Ancey** | Associate Professor, EPFL. PhD thesis advisor. Mail: christophe.ancey@epfl.ch
- **Mirko Musa** | Tenure Track Professor, EPFL. Postdoctoral PI. Mail: mirko.musa@epfl.ch